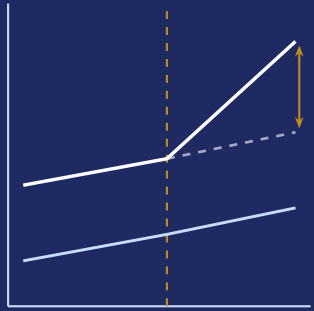


DIFFERENCES-IN-DIFFERENCES

Differences-in-Differences in Panel Models

Class 4 · Hour 1

Prof. Dr. Rodrigo Leite
Panel Data · EBAPE/FGV

CLASS 4 · HOUR 1

What We'll Cover This Hour

01 **The 2×2 DiD design**
Two groups, two periods, and the logic of the double difference

02 **Parallel trends and the other assumptions**
What they mean, why they matter, and how (far) you can check them

03 **DiD as a two-way fixed effects model**
Connecting today's tool to everything we've built so far

04 **Staggered adoption and its pitfalls**
Why DiD with multiple treatment dates needs extra care

05 **Estimating and validating DiD in Stata**
Event studies, placebo tests, clustered SEs, and `xtdidregress`

Recap: From Estimator Choice to Causal Design

So far: pooled OLS, FE, RE, formal tests, time effects, pseudo-panels, and IV — tools for estimating relationships as cleanly as the data allow.

Today we shift focus: not just estimating a relationship, but building a **comparison** — treated vs. control, before vs. after — that supports a causal claim.

Differences-in-differences is the workhorse of applied causal inference with panel data.

The Problem: Before/After Isn’t Enough

Before/after only

The treated units’ change over time also captures macro trends, seasonality, and any other shock hitting that period.

Treated vs. control only

A single cross-sectional comparison confuses the treatment’s effect with pre-existing differences between the groups.

DiD combines both comparisons, so each one cancels what the other cannot handle alone.

The Canonical 2×2 Design

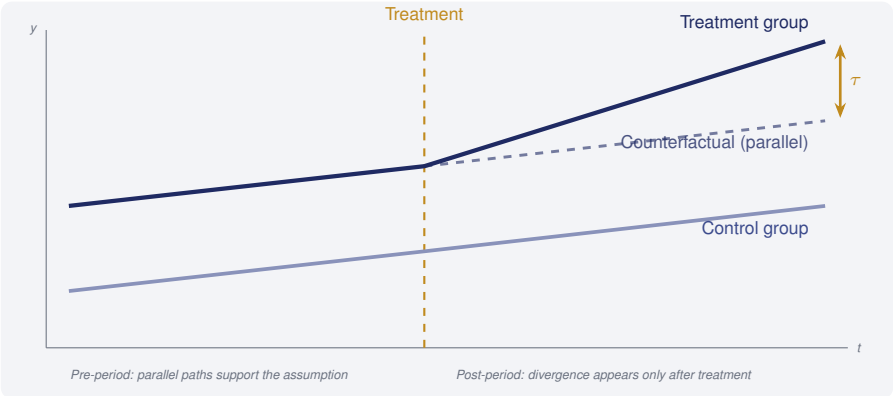
Two groups, two periods: the simplest version of the idea, and the building block for everything else today.

	Pre-period	Post-period
Control group	μ	$\mu + \lambda$
Treatment group	$\mu + \gamma$	$\mu + \gamma + \lambda + \boldsymbol{\tau}$

μ: baseline γ: permanent group gap λ: common time trend τ: treatment effect

$$\underbrace{\big[(\mu+\gamma+\lambda+\tau)-(\mu+\gamma)\big]}_{\text{treated change}} - \underbrace{\big[(\mu+\lambda)-\mu\big]}_{\text{control change}} = \tau$$

The Picture: Parallel Trends and the Counterfactual



The dashed line is never observed: it is the control group's change, added to the treated group's starting point.

The DiD Equation

$$y_{it} = \beta_0 + \beta_1 \, \text{Treat}_i + \beta_2 \, \text{Post}_t + \beta_3 \, (\text{Treat}_i \times \text{Post}_t) + \varepsilon_{it}$$

1	$\textbf{Treat}_i$	1 for units that ever receive treatment: captures the permanent group gap γ
2	$\textbf{Post}_t$	1 for periods after treatment starts: captures the common time change λ
3	$\textbf{Treat}_i \times \textbf{Post}_t$	1 only for treated units in the post-period
4	β_3	The DiD estimate of τ: causal under the assumptions on the next slides

Interpreting β_3: The Difference of Differences

"How much more did the treated group change, from before to after, than the control group changed over the same period?"

$$\hat{\beta}_3 = \left(\bar{y}_{T,\text{post}} - \bar{y}_{T,\text{pre}}\right) - \left(\bar{y}_{C,\text{post}} - \bar{y}_{C,\text{pre}}\right)$$

The control group's change is the counterfactual for what would have happened to the treated group without treatment.

What it estimates: the **ATT** — the average effect of treatment *on the treated*, not on the whole population.

The Parallel Trends Assumption

In the absence of treatment, the treated and control groups would have followed the **same trend** over time (not necessarily the same level).

1	**It is what makes β_3 causal**	Without it, the control group's change is not a valid counterfactual for the treated group
2	**Untestable in the post-period**	We never observe what the treated group would have done without treatment
3	**Partially checkable before treatment**	If trends already diverged before treatment, the assumption looks shaky
4	**Scale-dependent**	Parallel in levels does not imply parallel in logs, and vice versa: choose the outcome's scale deliberately

What Happens If Parallel Trends Fails

Trends genuinely parallel

β_3 recovers the causal effect: the control group's path is a valid stand-in for the counterfactual.

Trends already diverging

β_3 mixes the treatment effect with the pre-existing divergence, biased in whichever direction the trends were already moving.

This is why every serious DiD study shows a pre-trends (event-study) plot before reporting β_3.

Beyond Parallel Trends: The Other Assumptions

No anticipation

Units don’t react before treatment starts. If firms adjust once a law is announced, the “pre” period is already treated.

No spillovers (SUTVA)

Treatment of one unit doesn’t affect the controls. Competitors of treated firms, or neighboring municipalities, may be affected too.

Stable composition

The groups don’t change because of treatment: no selective entry, exit, or relocation in response to it.

And: controls should not themselves be affected by treatment. A “bad control” absorbs part of the effect you want to measure.

Testing Pre-Trends: The Event-Study Approach

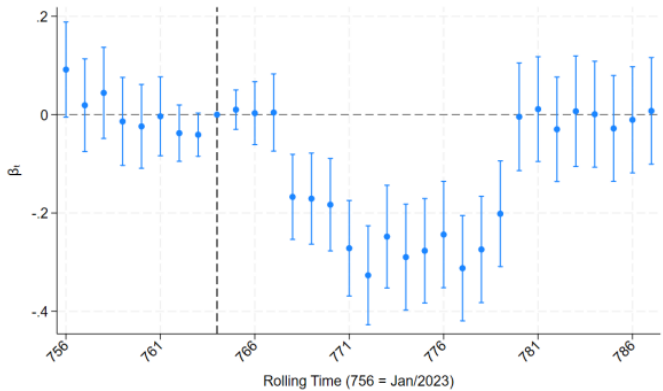
Replace the single Post indicator with one dummy per period *relative to treatment*, each interacted with Treat.

$$y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \delta_k \left(\text{Treat}_i \times \mathbf{1}[t - E_i = k] \right) + \varepsilon_{it}$$

E_i: treatment date of unit i; k: event time. With common timing, E_i is the same for every treated unit.

1. **Each δ_k is the gap between groups at event time k** — Relative to the omitted baseline, usually $k = -1$, the period just before treatment
2. **Pre-treatment δ_k should be near zero and flat** — The visual and statistical signature of parallel trends
3. **Post-treatment δ_k show the effect over time** — Whether it grows, fades, or stays constant

What an Event-Study Plot Looks Like



Coefficients $\hat{\delta}_k$ with 95% confidence intervals. Flat, insignificant pre-period estimates; a clear break after treatment.

Pre-Trend Tests: Useful, but Handle With Care

1	**"No significant pre-trend" $\neq$ "parallel trends"**	Pre-period tests often have low power: a real violation can easily go undetected
2	**Pre-testing distorts inference**	Reporting results only when the pre-trend test passes biases the estimates you then publish (Roth, 2022)
3	**Ask how big a violation would matter**	Sensitivity analysis bounds τ under violations as large as the observed pre-trends (Rambachan & Roth, 2023; `honestdid` in Stata)
4	**The argument comes first**	Explain *why* the groups should have trended together: institutional knowledge, not just a plot

DiD Is a Special Case of Two-Way Fixed Effects

Everything from Class 2 applies: DiD is what two-way FE looks like with a binary treatment.

1. **Treat_i is absorbed by α_i** — It is constant within each unit, so unit fixed effects already capture it
2. **Post_t is absorbed by λ_t** — It is common to all units in each period, so time fixed effects already capture it
3. **Only the interaction survives** — $y_{it} = \beta\, D_{it} + \alpha_i + \lambda_t + \varepsilon_{it}$, where $D_{it} = \text{Treat}_i \times \text{Post}_t$

> In a balanced 2×2 panel without covariates, this TWFE regression reproduces the hand-computed DiD estimate *exactly*. With covariates or many periods, it is a regression-weighted version of it.

Beyond 2×2: Multiple Periods and Staggered Treatment

Real panels rarely offer a single treatment date: different units often adopt treatment at different times.

Common timing

Every treated unit starts treatment in the same period: the 2×2 logic generalizes cleanly.

Staggered timing

Units adopt treatment in different periods — states passing a law in different years, firms adopting Pix at different dates.

The natural approach is to run the same TWFE regression with staggered timing. It looks like a harmless generalization — **it isn't**.

The Staggered Adoption Problem

With staggered timing, TWFE implicitly uses **already-treated units as controls** for units treated later.

1	**Those comparisons are contaminated**	The already-treated unit's own (changing) treatment effect leaks into the "control" trend
2	**Some effects get negative weight**	If effects vary over time or across cohorts, the TWFE estimate can be badly distorted — even of the wrong sign
3	**A structural problem**	It comes from the estimator itself, not from small samples — and it disappears if the effect is constant across units and time

The Goodman-Bacon Decomposition

Goodman-Bacon (2021): the TWFE estimate with staggered timing is a weighted average of every possible 2×2 comparison in the data.

1	**Treated vs. never-treated**	Clean comparisons that behave like a standard 2×2 DiD
2	**Earlier- vs. later-treated, *before* the later group is treated**	Also fine: the later group is still untreated, a valid control under parallel trends
3	**Later- vs. earlier-treated, *after* the earlier group is treated**	The problematic "forbidden" comparisons: the control group is already treated
4	**Weights depend on group sizes and timing**	Groups treated near the middle of the panel get the most weight, because their treatment status varies most

In Stata: `bacondecomp` (user-written) shows each comparison's estimate and weight.

Modern Solutions: Heterogeneity-Robust Estimators

1	**Callaway & Sant'Anna (2021)**	Estimates group-time ATTs using only clean controls (never- or not-yet-treated), then aggregates them
2	**Sun & Abraham (2021)**	An interaction-weighted event-study estimator that avoids already-treated units as controls
3	**de Chaisemartin & D'Haultfœuille (2020)**	Diagnoses negative weights and offers an estimator robust to them
4	**Borusyak, Jaravel & Spiess (2024)**	Imputation: fit FE on untreated observations only, then impute the counterfactuals
5	**Practical takeaway**	With staggered timing, don't default to plain TWFE: run the diagnostics and report a robust estimator

Staggered DiD in Stata: The Main Tools

```
ssc install bacondecomp          // Goodman-Bacon decomposition
ssc install drdid
ssc install csdid                // Callaway & Sant'Anna
ssc install eventstudyinteract   // Sun & Abraham
ssc install did_multiplegt       // de Chaisemartin & D'Haultfoeuille
ssc install did_imputation       // Borusyak, Jaravel & Spiess

* Callaway & Sant'Anna: gvar = first treatment period (0 = never treated)
csdid y x, ivar(id) time(year) gvar(first_treat) notyet
estat event                      // event-study aggregation
```

Stata 18+ also ships `xthdidregress` for heterogeneous-effect DiD. Check each package's help file: several need companion packages.

Standard Errors: Serial Correlation

Bertrand, Duflo & Mullainathan (2004): DiD standard errors that ignore serial correlation can drastically understate uncertainty.

1	**Outcomes are serially correlated**	The same issue we met with pooled OLS in Class 1
2	**DiD makes it worse**	The treatment indicator is itself extremely persistent: once on, it stays on
3	**Cluster at the level of assignment**	By unit — or by state, if a state law assigns treatment to all firms in that state
4	**Few treated clusters need care**	With a handful of treated units, use the wild cluster bootstrap or randomization inference

Robustness Checks: Placebo Tests

1	**Placebo treatment timing**	Pretend treatment happened earlier than it did: no “effect” should appear before the true date
2	**Placebo treatment groups**	Assign “treatment” to units never actually treated: the estimate should be indistinguishable from zero
3	**Placebo outcomes**	Use an outcome the treatment should not affect: a significant “effect” signals confounding
4	**Alternative control groups**	Results should hold with a different, plausible comparison group
5	**Leads-and-lags event study**	The single most expected robustness check in any DiD paper

Estimating DiD in Stata (1/2): TWFE

```
clear all
use https://www.stata-press.com/data/r19/patents
xtset classid year
bysort classid: egen byte treat = max(gotpatent)   // ever-treated class
generate byte post = year > 1918                   // after the 1919 licensing
generate byte did  = treat * post
xtreg uspatents did fpatents i.year, fe vce(cluster classid)
```

`fe`	unit FE absorb `treat`
`i.year`	time FE absorb `post`
`did`	$\text{Treat}_i \times \text{Post}_t$: its coefficient is the DiD estimate
`vce(cluster classid)`	clustered SEs at the level of treatment assignment

Data: Stata's example on compulsory licensing of patents in 1919 (Moser & Voena, 2012).

Estimating DiD in Stata (2/2): Event Study and Pre-Trends

```
* Event study: one treat x year interaction per year, 1918 as baseline
xtreg uspatents ib1918.year#c.treat fpatents i.year, fe vce(cluster classid)
* (plot the interactions with coefplot: ssc install coefplot)

* Differential linear pre-trend, pre-period only
xtreg uspatents c.year#c.treat fpatents i.year if year < 1919, ///
      fe vce(cluster classid)
```

`ib1918.year#c.treat`	the δ_k: gap between groups in each year, relative to 1918
`c.year#c.treat`	tests whether treated classes were already trending differently

Testing `treat` itself before treatment (e.g., with RE) only checks a *level* difference, which DiD allows. Parallel trends is about *slopes*.

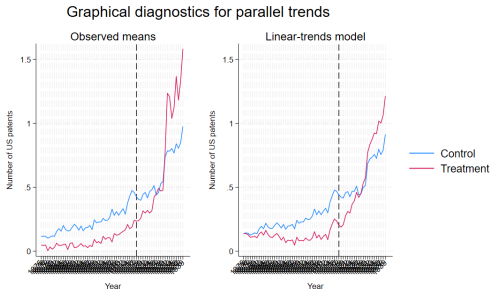
Built-In DiD in Stata (17+): `xtdidregress`

```
xtdidregress (uspatents fpatents) (did), group(classid) time(year)
estat ptrends        // test of parallel linear pre-trends
estat granger        // anticipation test: "effects" before treatment?
estat trendplots     // observed means and linear-trends model
estat grangerplot    // leads and lags, event-study style
```

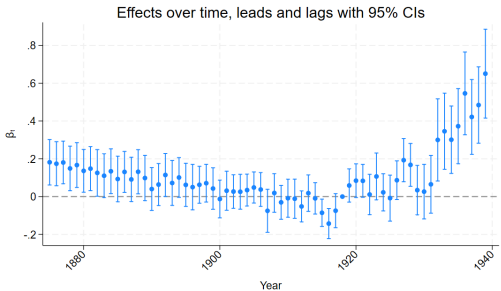
`xtdidregress`	DiD with unit and time FE; SEs clustered at the `group()` level by default
`estat ptrends`	H_0: treated and control had the same linear trend before treatment
`estat granger`	H_0: no anticipation — no treatment "effects" in the leads

`xtdidregress` assumes a common treatment time or a constant effect; for staggered designs see `xthdidregress` (Stata 18+).

`estat trendplots` and `estat grangerplot`



Observed means and linear-trends model



Leads and lags with 95% CIs

In class: do the pre-1919 leads in the right-hand plot support parallel trends? What would you say about the post-1919 path?

Check Your Understanding

1. **Which assumption?** A minimum-wage increase is announced a year before it takes effect. Which DiD assumption is at risk, and how would you see it in an event study?
2. **Pre-trends** Your pre-period leads are insignificant but steadily rising. Is parallel trends supported?
3. **Staggered timing** States adopt a law between 2010 and 2018 and effects grow over time. Why is plain TWFE risky here?
4. **Clustering** Treatment is assigned at the state level; your data are firm-years. At which level do you cluster?

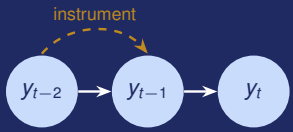
Recap and What's Next

Takeaway: DiD turns panel structure into a causal research design — powerful under parallel trends, but requiring real care with pre-trends, standard errors, and staggered treatment timing.

Next: Hour 2 — Dynamic Models: Autocorrelation & GMM

What happens when y depends on its own past, and how lagged values become instruments.

Questions?



DYNAMIC PANEL MODELS

Dynamic Models: Autocorrelation & GMM

Class 4 · Hour 2

Prof. Dr. Rodrigo Leite
Panel Data · EBAPE/FGV

CLASS 4 · HOUR 2

What We'll Cover This Hour

01 **The dynamic panel model**
When and why a lagged dependent variable belongs on the right-hand side

02 **Dynamic panel bias (Nickell bias)**
Why fixed effects fails once y is its own regressor

03 **Autocorrelation in panel errors**
A related but distinct problem, and how to test for it

04 **From Anderson–Hsiao to GMM**
Building up to Arellano–Bond and System GMM step by step

05 **Diagnostics and Stata**
AR tests, Hansen J, instrument counts, and `xtabond2`

Recap: A Loose End From Class 3

We flagged it before: when the lagged dependent variable is a regressor, fixed effects stops being consistent — the Nickell bias.

Today we unpack why that happens, and build the standard toolkit researchers use to fix it.

We also revisit autocorrelation: a related, but distinct, problem in panel errors — and one that interacts with the dynamic model in an important way.

The Dynamic Panel Model

$$y_{it} = \rho\, y_{i,t-1} + \beta x_{it} + \alpha_i + \varepsilon_{it}, \qquad |\rho| < 1$$

1	$y_{i,t-1}$ **is now a regressor**	Last period's outcome helps explain this period's outcome
2	ρ **measures persistence**	How much of last period's value carries over into this one
3	**Short- and long-run effects differ**	β is the immediate effect of x; $\beta/(1-\rho)$ is the long-run effect (Class 1, Hour 2)
4	**Not a niche case**	Profitability, capital structure, sales, prices, and wages are all commonly modeled this way

Why Include a Lagged Dependent Variable?

Persistence and habit formation

Consumption, routines, and organizational practices depend directly on their own recent past.

Partial adjustment

Firms rarely jump to a new target capital structure or staffing level; they adjust gradually, period by period.

State dependence

In labor economics, past unemployment can causally raise the probability of future unemployment.

Better-specified dynamics

Omitting a truly dynamic process pushes it into the error term, which becomes serially correlated.

The Problem: Dynamic Panel Bias

Nickell (1981): in a dynamic panel, the within (FE) estimator of ρ is inconsistent for fixed T, even as $N \to \infty$.

$$\text{plim}_{N\to\infty}\left(\hat{\rho}^{FE} - \rho\right) \approx -\frac{1+\rho}{T-1} \qquad \text{(for moderately large } T\text{)}$$

1. **Usually negative** — For $\rho > 0$, FE underestimates persistence
2. **Of order $1/T$** — $T = 10$, $\rho = 0.5$: bias ≈ -0.17, painful in short panels and minor in long ones
3. **More units don't help** — Larger N only makes the estimate more precisely wrong; only larger T helps

Why Fixed Effects Fails Here, Mechanically

The within transformation subtracts each unit's mean from every variable:

$$(y_{it} - \bar{y}_i) = \rho\,(y_{i,t-1} - \bar{y}_{i,-1}) + (\varepsilon_{it} - \bar{\varepsilon}_i)$$

1. **$\bar{\varepsilon}_i$ contains $\varepsilon_{i,t-1}$** — And $y_{i,t-1}$ depends on $\varepsilon_{i,t-1}$, so the demeaned regressor and the demeaned error are correlated
2. **$\bar{y}_{i,-1}$ contains future y's** — Which depend on ε_{it} — a second channel of correlation
3. **A textbook exogeneity violation** — Each term is weighted by $1/T$, which is why the bias shrinks only as T grows

A Related but Different Issue: Autocorrelated Errors

Dynamic panel bias

- Comes from including $y_{i,t-1}$ with FE
- Biases the coefficient estimate itself
- Needs a different estimator (IV/GMM)

Autocorrelated errors

- Comes from ε_{it} correlated with $\varepsilon_{i,t-1}$
- In a *static* model: coefficients consistent, SEs wrong
- Fixed with clustered SEs (or an AR(1) error model)

The two interact. With $y_{i,t-1}$ in the model, serially correlated ε_{it} makes $y_{i,t-1}$ correlated with ε_{it} — and it invalidates the shallowest lags as GMM instruments. That is what the AR(2) test checks.

Testing for Autocorrelation: The Wooldridge Test

Wooldridge's test uses the residuals from the first-differenced (static) regression.

> If ε_{it} has no serial correlation, then $\text{Corr}(\Delta\varepsilon_{it}, \Delta\varepsilon_{i,t-1}) = -0.5$. The test regresses the FD residuals on their lag and tests whether the coefficient equals -0.5.

1	**H_0: no first-order serial correlation**	In the idiosyncratic errors ε_{it}; the test is robust to heteroskedasticity
2	**Reject H_0**	Autocorrelation is present: classical SEs are unreliable, and in a dynamic model lag choices for GMM matter
3	**In Stata:** `xtserial`	User-written (Drukker, 2003); find it with `search xtserial`

The First Step: First-Differencing

Instead of demeaning, take first differences: the unit effect still cancels, but the correlation problem changes shape.

$$\Delta y_{it} = \rho \, \Delta y_{i,t-1} + \beta \, \Delta x_{it} + \Delta \varepsilon_{it}$$

1	α_i **is gone**	Differencing removes it, just as demeaning does
2	$\Delta y_{i,t-1}$ **still correlates with** $\Delta \varepsilon_{it}$	$\Delta y_{i,t-1} = y_{i,t-1} - y_{i,t-2}$ contains $y_{i,t-1}$, which depends on $\varepsilon_{i,t-1}$ — and $\Delta \varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$
3	**But the correlation is now local**	Only $\varepsilon_{i,t-1}$ links them, so anything dated $t-2$ or earlier is a candidate instrument

The Anderson–Hsiao Estimator

Anderson & Hsiao (1981): instrument $\Delta y_{i,t-1}$ with $y_{i,t-2}$ in levels (or with $\Delta y_{i,t-2}$).

1	**Relevance**	$y_{i,t-2}$ is correlated with $\Delta y_{i,t-1} = y_{i,t-1} - y_{i,t-2}$
2	**Exogeneity**	$y_{i,t-2}$ is uncorrelated with $\Delta \varepsilon_{it}$, provided the ε_{it} are not serially correlated
3	**A just-identified IV**	Consistent, estimated by 2SLS — but it leaves useful information unused

The Limitation: Information Left on the Table

Once $T > 3$, more valid instruments become available — and Anderson–Hsiao ignores them.

1. **One instrument per equation** — $y_{i,t-2}$ alone, when $y_{i,t-3}$, $y_{i,t-4}$, and earlier lags are also valid
2. **Inefficient in practice** — Especially when the extra lags carry real information about $\Delta y_{i,t-1}$
3. **The valid set grows with t** — Later periods have more legitimate instruments; a single fixed choice wastes them

> This is exactly the gap that GMM was designed to close.

The GMM Idea: Use Every Valid Lag

Each period gets its own, growing set of instruments, expressed as moment conditions:

$$E\left(y_{i,t-s}\,\Delta\varepsilon_{it}\right) = 0 \qquad \text{for } s \geq 2, \quad t = 3, \ldots, T$$

Eq. for	Instruments	#
$t = 3$	y_{i1}	1
$t = 4$	y_{i1}, y_{i2}	2
$t = 5$	y_{i1}, y_{i2}, y_{i3}	3
$\vdots$	$\vdots$	$\vdots$
$t = T$	$y_{i1}, \ldots, y_{i,T-2}$	$T - 2$

The estimator combines all these conditions, weighting them by their estimated covariance.

Total instruments for y alone: $(T-1)(T-2)/2$ — growing **quadratically** in T.

Arellano–Bond: Difference GMM

Arellano & Bond (1991): estimate the first-differenced equation by GMM, using all available lagged levels of y (and of x, if not strictly exogenous) as instruments.

1. **Instruments grow with t** — For period t: $y_{i,t-2}, y_{i,t-3}, \ldots, y_{i1}$ — the key advance over Anderson–Hsiao
2. **Works when persistence is moderate** — Lagged levels must be informative about future changes
3. **Known weakness: high persistence** — When $\rho \to 1$ (or σ_α^2 is large), lagged levels say little about Δy: weak instruments

How to Treat the Other Regressors

In the differenced equation, the valid lags of x depend on how x relates to ε:

Type of x_{it}	Assumption	xtabond2
Strictly exogenous	Uncorrelated with ε_{is} in all periods	`iv(x)`
Predetermined	Uncorrelated with current and future shocks; may respond to past ones	`gmm(x, lag(1 .))`
Endogenous	Correlated with the current shock	`gmm(x, lag(2 .))`

Treating x as exogenous when it isn't is one of the most common errors in applied GMM.

System GMM: Arellano–Bover / Blundell–Bond

The fix for weak instruments under high persistence: add the levels equation, instrumented by lagged *differences*, and estimate both as a system.

1	**Two equations, two instrument sets**	Differenced equation: lagged levels. Levels equation: lagged differences ($\Delta y_{i,t-1}$)
2	**More efficient for persistent series**	The levels moment conditions add real information when ρ is high
3	**An extra assumption**	$E(\Delta y_{i,t-1}\,\alpha_i) = 0$: deviations from the long-run mean are unrelated to α_i (mean stationarity of initial conditions) — plausible in many, not all, settings
4	**Common in corporate finance**	Especially with short panels and persistent variables like leverage or cash holdings

One-Step vs. Two-Step GMM

	One-step	Two-step
Weighting matrix	Simple, pre-specified	Re-estimated from first-step residuals
Efficiency	Lower	Asymptotically efficient
Standard errors	Reliable (use robust)	Severely downward biased in finite samples — unless Windmeijer-corrected
Typical use	A conservative default	With `robust`: two-step plus the Windmeijer (2005) correction

In `xtabond2`, `twostep robust` applies the Windmeijer correction automatically.

Diagnostics You Must Report

1	**AR(1) in differenced residuals**	Expected to reject: $\Delta\varepsilon_{it}$ and $\Delta\varepsilon_{i,t-1}$ share $\varepsilon_{i,t-1}$ by construction
2	**AR(2) in differenced residuals**	Should **not** reject. If it does, ε_{it} is serially correlated and $y_{i,t-2}$ is not a valid instrument: start from deeper lags
3	**Hansen *J* test**	Joint validity of the overidentifying restrictions (robust; prefer it to Sargan). A high *p*-value is reassuring, not proof
4	**Difference-in-Hansen**	For System GMM: tests the extra levels instruments specifically
5	**The instrument count**	Always report it next to the number of groups

The Instrument Proliferation Problem

The instrument count grows quadratically in T. You can easily end up with more instruments than units.

1	**Overfitting the endogenous variables**	Too many instruments pull GMM back toward the biased OLS/FE estimate
2	**A weakened Hansen test**	With many instruments, J loses power and can show an implausible $p = 1.000$
3	**The fix: collapse or limit lags**	`collapse` and `lag(2 3)`-style limits in `xtabond2`; check that results are stable across choices

> Rule of thumb (Roodman, 2009): keep the number of instruments well below the number of groups, and report both.

A Sanity Check: The OLS–FE Bounds

Pooled OLS: biased up

$y_{i,t-1}$ contains α_i, which is also in the error: $\hat{\rho}^{OLS}$ is biased upward

FE: biased down

The Nickell bias pushes $\hat{\rho}^{FE}$ downward (for $\rho > 0$)

GMM: in between

A credible $\hat{\rho}^{GMM}$ should lie between the two. If it doesn't, suspect weak or invalid instruments

$$\hat{\rho}^{FE} \lesssim \hat{\rho}^{GMM} \lesssim \hat{\rho}^{OLS}$$ (Bond, 2002)

Dynamic GMM in Stata (1/2): Difference vs. System

```
ssc install xtabond2
clear all
use https://www.stata-press.com/data/r19/patents
xtset classid year
quietly tabulate year, generate(yr_)
local window "if year > 1900 & year < 1919"      // pre-treatment period
* Difference GMM (Arellano-Bond): noleveleq
xtabond2 uspatents L.uspatents fpatents yr_* `window', ///
    gmm(uspatents, lag(2 3) collapse) iv(fpatents yr_*) noleveleq twostep robust artests(3)
* System GMM (Blundell-Bond): the default in xtabond2
xtabond2 uspatents L.uspatents fpatents yr_* `window', ///
    gmm(uspatents, lag(2 3) collapse) iv(fpatents yr_*) twostep robust artests(3)
```

`gmm(uspatents, lag(2 3) collapse)`	instruments: y_{t-2} and y_{t-3}, collapsed; `iv()`: strictly exogenous regressors
`noleveleq / twostep robust`	Difference GMM / two-step with Windmeijer-corrected SEs

Dynamic GMM in Stata (2/2): Endogenous x, Bounds

```
* fpatents treated as endogenous: instrument it with its own lags 2-3
xtabond2 uspatents L.uspatents fpatents yr_* `window', ///
    gmm(uspatents fpatents, lag(2 3) collapse) iv(yr_*) twostep robust artests(3)

* Bounds check for rho: pooled OLS (upper) and FE (lower)
regress uspatents L.uspatents fpatents yr_* `window', vce(cluster classid)
xtreg   uspatents L.uspatents fpatents yr_* `window', fe vce(cluster classid)
```

`gmm(...fpatents, lag(2 3))` endogenous x; use `lag(1 3)` if it is only predetermined

`regress / xtreg, fe` the coefficient on `L.uspatents` should bracket the GMM estimate

Run all the lines in one go: `` `window' `` is a local macro and only exists while the do-file runs.

Reading the `xtabond2` Output

Number of instruments	Compare with the number of groups: it should be clearly smaller
Arellano–Bond AR(1)	Expect a small p-value
Arellano–Bond AR(2)	Expect a large p-value; if not, start the lags deeper
Hansen test	Want a moderate p-value; a p near 1.000 signals too many instruments
Difference-in-Hansen	System GMM: validity of the levels-equation instruments

In class: is $\hat{\rho}^{GMM}$ between the OLS and FE estimates? Do the results survive `lag(2 4)` instead of `lag(2 3)`?

Check Your Understanding

1	**Nickell bias**	You have $N = 3{,}000$ firms and $T = 6$ years. Will more firms fix the FE estimate of ρ?
2	**AR(2)**	Your AR(2) test rejects with $p = 0.01$. What does it mean, and what do you change?
3	**Hansen $p = 1.000$**	With 250 instruments and 180 firms, should you be reassured?
4	**Which GMM?**	Leverage is extremely persistent ($\hat{\rho}^{OLS} \approx 0.95$). Difference or System GMM, and what extra assumption do you accept?

Recap and What's Next

Takeaway: once y depends on its own past, fixed effects is no longer enough. Arellano–Bond and System GMM extend the IV logic from Class 3 into a toolkit for persistence — with diagnostics you must respect.

Next: Hour 3 — Paper Discussions

Applied papers. The complete toolkit: pooled OLS, FE/RE, time effects, pseudo-panels, IV, DiD, and dynamic GMM — you're ready for the exam!

Questions?

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References (2/2): Dynamic Panels

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